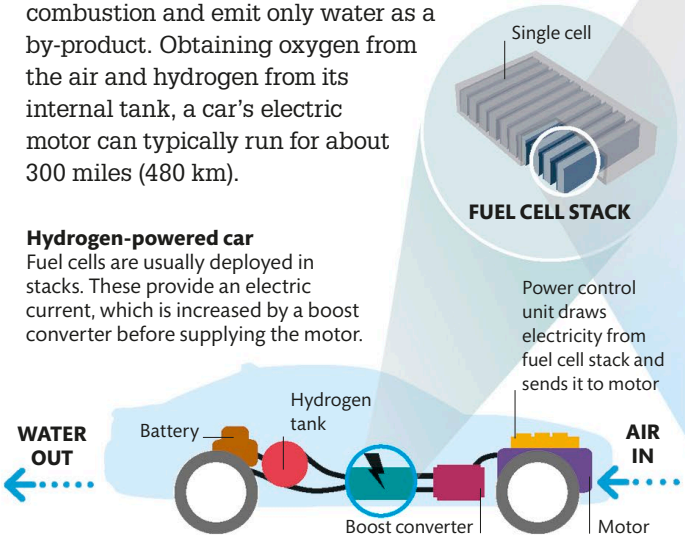
Fuel cells

Fuel cells generate electricity through a chemical reaction caused by mixing fuel with oxygen. There are a number of types, but cells using hydrogen are increasingly used in vehicles and electronic devices.

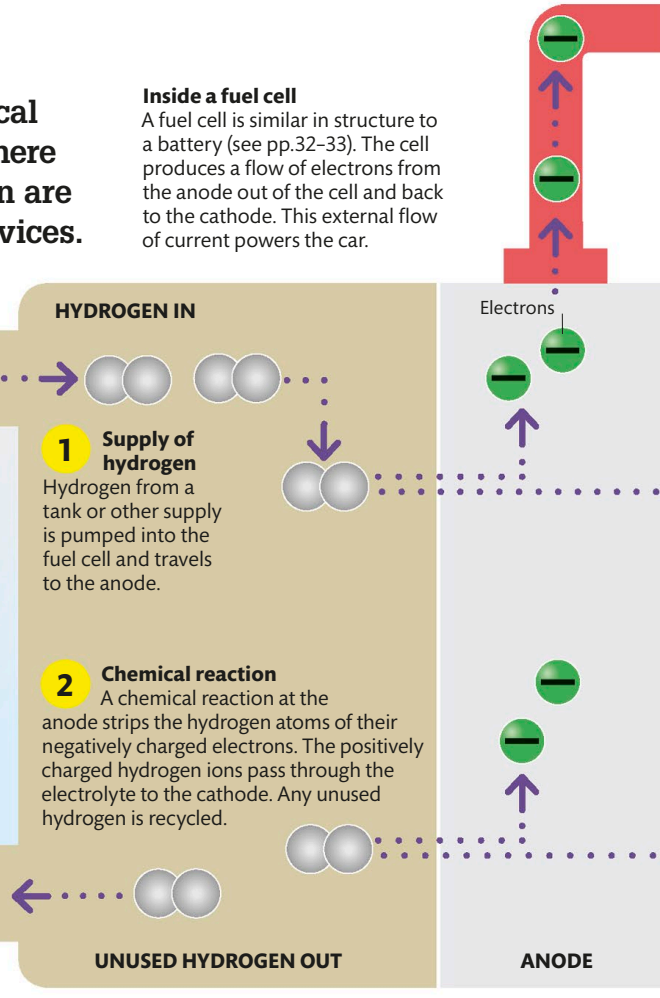
How a fuel cell works

A fuel cell is an electrochemical cell that produces an electric current, which is used to power motors or other electrical devices. Hydrogen fuel cells produce electricity without combustion and emit only water as a by-product. Obtaining oxygen from the air and hydrogen from its internal tank, a car's electric motor can typically run for about 300 miles (480 km).

Hydrogen-powered car
Fuel cells are usually deployed in stacks. These provide an electric current, which is increased by a boost converter before supplying the motor.



Inside a fuel cell
A fuel cell is similar in structure to a battery (see pp.32–33). The cell produces a flow of electrons from the anode out of the cell and back to the cathode. This external flow of current powers the car.



Sources of hydrogen

Most hydrogen is produced from fossil fuels, particularly natural gas. The most common method used is a process called steam-methane reformation, which produces some carbon dioxide emissions. Other processes, such as electrolysis, harness hydrogen without harmful emissions but use lots of energy.

Steam-methane reformation
Methane and steam react to produce a mixture of gases, which are sent to a shift reactor where more hydrogen and carbon dioxide are created. A purification stage produces pure hydrogen.

